

Green Reciprocity for Globally Consistent Normal Orientation

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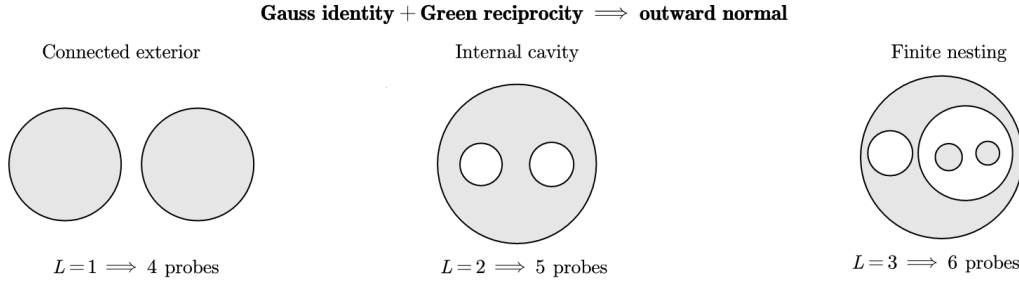


Fig. 1. A continuous well-posedness theory for global normal orientation. Full Green reciprocity gives topology-independent identifiability. A geometry of maximum nesting depth L needs only $L + 3$ fixed probes to reconstruct globally consistent outward normal.

Globally consistent normal orientation is a fundamental problem in geometry processing. Local geometry determines a normal line, but not which of its two directions is globally outward. We study the continuous inverse problem underlying this ambiguity. Our formulation combines the Gauss formula with a weak antisymmetric Green-reciprocity condition. We establish a well-posedness theory with four components. First, for every bounded smooth open set of finite topology—including disconnected objects, internal cavities, and arbitrary finite nesting—the full formulation has the outward unit normal as its unique smooth solution. Second, the infinite reciprocity family admits a topology-aware finite compression. If the maximum boundary nesting depth is L , then L threshold probes and the three coordinate probes suffice to solve the outward normal. Third, for perturbed Gauss and reciprocity data we characterize exact solvability by the range of an explicit linear boundary operator coupling the normal coefficient and the tangential field. The same $L + 3$ slices reconstruct the only possible candidate solution, while the remaining reciprocity data provide a necessary and sufficient consistency test. Fourth, on the admissible data range the inverse is geometry-dependent Lipschitz continuous in Sobolev norms.

Additional Key Words and Phrases: normal orientation, Green reciprocity, layer potentials, geometric analysis, identifiability, solvability, stability

1 Introduction

A smooth surface determines a normal line locally, but not a globally consistent choice between its two directions. This distinction is central to geometry processing. Local neighborhoods can provide reliable unoriented normal directions, while their signs must be coordinated over an entire shape or scene.

Classical methods propagate signs over neighborhood graphs, beginning with the minimum-spanning-tree strategy of Hoppe et al. [1992]. Global implicit and potential-theoretic formulations, notably Poisson surface reconstruction [Kazhdan et al. 2006] and generalized winding numbers [Jacobson et al. 2013], provide important background for later orientation methods. Orientation-specific approaches include parametrized Gauss reconstruction [Lin et al. 2022], regularized winding-number fields [Xu et al. 2023], Poisson-isovalue constraints [Xiao et al. 2023], boundary integration [Liu et al. 2024], Stokes-based linear constraints [Gotsman and Hormann 2024], stochastic orientation objectives [Huang

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et al. 2024], winding-number consistency [Lin et al. 2024], Gauss-based extensions [Ma et al. 2025a,b], and Delaunay-graph least squares [Fu et al. 2025]. Learning-based methods improve either local normal estimation [Ben-Shabat and Gould 2020; Guerrero et al. 2018; Zhu et al. 2021] or oriented prediction itself [Li et al. 2023, 2024; Metzger et al. 2021; Wang et al. 2022].

These developments leave a more basic question that is independent of any particular discretization:

Can we formulate a mathematical model to uniquely solve the outward normal, how much information is needed, and how stable is the inverse?

We answer these questions using a weak antisymmetric identity induced by Green’s second identity, which we call *Green reciprocity*. Combined with the Gauss formula, it defines a linear continuous inverse problem for an unconstrained boundary vector field. The central decomposition

$$m = av + \tau, \quad \tau \cdot v = 0, \quad v \text{ is the outward normal,}$$

separates a scalar normal-coefficient ambiguity from a tangential ambiguity.

The first result is topology-independent:

$$\text{Gauss identity} + \text{full Green reciprocity} \implies m = v.$$

It holds for arbitrary bounded smooth finite topology, without connectedness assumptions on the region or its complement.

The second result quantifies the information contained in full reciprocity. Root the region-adjacency tree at the unbounded exterior and let L be the maximum boundary nesting depth. We construct threshold functions $\chi_1, \dots, \chi_L$ whose harmonic extensions have strict alternating Hopf signs on successive boundary layers. These probes determine the normal coefficient one layer at a time. The coordinate functions x_1, x_2, x_3 then eliminate the tangential field. Hence $L + 3$ fixed probe families suffice. Connected-exterior geometries require four probes, and ordinary cavity geometries require five.

The third and fourth results concern perturbed data. After conjugating the reciprocity datum by the single-layer operator, the perturbation is represented by a boundary operator. Exact solvability is characterized as a range condition rather than merely a small-residual condition. The same $L + 3$ slices reconstruct the only possible pair (b, τ) . Full reciprocity then becomes an exact consistency test. On the admissible range, the inverse satisfies a geometry-dependent Lipschitz estimate.

The paper therefore studies one continuous inverse problem from four complementary viewpoints:

- **Identifiability:** the homogeneous null space is $\{0\}$.
- **Finite information:** $L + 3$ fixed probes determine the unique solution.
- **Exact solvability:** perturbed data are realizable precisely when they lie in the range of an explicit boundary operator.
- **Stability:** on that range, the inverse is Lipschitz continuous in the Sobolev norms.

2 Related Work

Local normals and sign propagation. The classical point-cloud pipeline estimates an unoriented normal in a local neighborhood and then propagates its sign through a graph [Hoppe et al. 1992]. Learning-based local estimators such

as PCPNet, DeepFit, and AdaFit substantially improve the first stage [Ben-Shabat and Gould 2020; Guerrero et al. 2018; Zhu et al. 2021], but local accuracy and global sign consistency remain logically distinct problems.

Global orientation. Generalized winding numbers [Jacobson et al. 2013] provide a foundational inside–outside representation that underlies several later orientation objectives. Orientation-specific global methods include parametrized Gauss reconstruction [Lin et al. 2022], winding-number regularization [Xu et al. 2023], boundary integration [Liu et al. 2024], Stokes constraints [Gotsman and Hormann 2024], stochastic normal orientation [Huang et al. 2024], Delaunay-graph least squares [Fu et al. 2025], and boundary-aware or Gauss-based variants [Ge et al. 2026; Ma et al. 2025a,b]. Learning-based orientation methods use learned patch or shape-level cues [Li et al. 2023, 2024; Metzger et al. 2021; Wang et al. 2022]. These works primarily address how to compute a consistent orientation; our question is whether a continuous global system identifies the target uniquely.

Green identities and integral formulations. Potential-theoretic ideas are already central to graphics formulations based on Gauss, winding numbers, and boundary integrals. G3R [Wu et al. 2026] explicitly uses Green’s third identity and analyzes the low-order harmonic family $\{1, x_1, x_2, x_3\}$. Our formulation instead starts from the antisymmetric bilinear relation induced by Green’s second identity. This leads to a different constraint system before the normal coefficient is known, and to a continuum uniqueness theorem over arbitrary finite topology. Standard layer-potential facts used below can be found in McLean [2000], maximum-principle and Hopf-lemma arguments follow Gilbarg and Trudinger [2001].

3 Problem Setup and Green-Reciprocity Formulation

3.1 Geometry and layer potentials

Let $\Omega \subset \mathbb{R}^3$ be a bounded open set whose boundary

$$\Sigma := \partial\Omega$$

is a finite disjoint union of closed, connected, C^∞ embedded surfaces. We call this *bounded smooth finite topology*. Neither Ω nor $\mathbb{R}^3 \setminus \overline{\Omega}$ is assumed connected. Let ν be the outward unit normal of Ω .

We use the fundamental solution

$$G(x, y) = \frac{1}{4\pi|x - y|}, \quad -\Delta_y G(x, y) = \delta_x(y).$$

For a smooth density ρ , define the single-layer potential and its boundary trace by

$$(S\rho)(z) = \int_{\Sigma} G(z, y)\rho(y) dS_y, \quad z \in \mathbb{R}^3 \setminus \Sigma,$$

$$(V\rho)(x) = \int_{\Sigma} G(x, y)\rho(y) dS_y, \quad x \in \Sigma.$$

Define the Neumann–Poincaré operators by

$$(K\varphi)(x) = \text{p. v.} \int_{\Sigma} \partial_{\nu_y} G(x, y)\varphi(y) dS_y,$$

$$(K'\rho)(x) = \text{p. v.} \int_{\Sigma} \partial_{\nu_x} G(x, y)\rho(y) dS_y,$$

where p. v. denotes the principal value.

On smooth closed surfaces in three dimensions, V is a self-adjoint elliptic isomorphism between the standard trace spaces, the Plemelj identity $KV = VK'$ holds, and

$$A := V^{-1}K = K'V^{-1}$$

is an $L^2(\Sigma)$ -self-adjoint classical pseudodifferential operator of order zero [Ando et al. 2025; McLean 2000]. We write $\nabla^{\text{p.v.}}V\rho$ for the arithmetic mean of the two one-sided boundary traces of $\nabla S\rho$. If $u = V\rho$, then

$$\nabla^{\text{p.v.}}V\rho = \nabla_\Sigma u + \nu Au. \quad (1)$$

At each point of Σ , the superscripts i and e below denote traces from the Ω side and from the $\mathbb{R}^3 \setminus \overline{\Omega}$ side, respectively. We use the one-sided normal-trace convention

$$\partial_\nu^i S\rho = K'\rho + \frac{1}{2}\rho, \quad \partial_\nu^e S\rho = K'\rho - \frac{1}{2}\rho, \quad (2)$$

so that

$$\partial_\nu^e S\rho - \partial_\nu^i S\rho = -\rho. \quad (3)$$

Thus $K'\rho$ is exactly the average of the two one-sided normal traces. These sign conventions are used repeatedly in the Hopf-lemma arguments below. The technical functional-analytic statements are collected in Appendix A.

3.2 Gauss and full Green reciprocity

The two constraints used below come directly from classical Green identities. For fixed $x \notin \Sigma$, the divergence theorem gives

$$\int_\Sigma \nabla_y G(x, y) \cdot \nu(y) dS_y = \begin{cases} -1, & x \in \Omega, \\ 0, & x \notin \overline{\Omega}. \end{cases} \quad (4)$$

Taking the standard principal-value trace on the boundary yields

$$\text{p. v.} \int_\Sigma \nabla_y G(x, y) \cdot \nu(y) dS_y = -\frac{1}{2}, \quad x \in \Sigma. \quad (5)$$

The reciprocity condition is motivated by Green's second identity. Let $x, x' \in \mathbb{R}^3 \setminus \overline{\Omega}$ and set

$$u(y) = G(x, y), \quad v(y) = G(x', y).$$

Both functions are harmonic in every connected component of Ω , so summing Green's second identity over those components gives

$$\int_\Sigma \left[G(x, y) \nabla_y G(x', y) - G(x', y) \nabla_y G(x, y) \right] \cdot \nu(y) dS_y = 0. \quad (6)$$

Formally moving both source points to Σ suggests a two-kernel boundary relation. Because this pointwise expression contains coupled singularities on the diagonal, we do not use it as a constraint. Instead, we test the two source variables against smooth densities ρ and η , integrate in the source variables first, and obtain the weak antisymmetric relation

$$\int_\Sigma [(V\rho) \nu \cdot \nabla^{\text{p.v.}}V\eta - (V\eta) \nu \cdot \nabla^{\text{p.v.}}V\rho] dS = 0. \quad (7)$$

This is the exact form that remains meaningful for arbitrary smooth test densities and motivates the definition of B_m below.

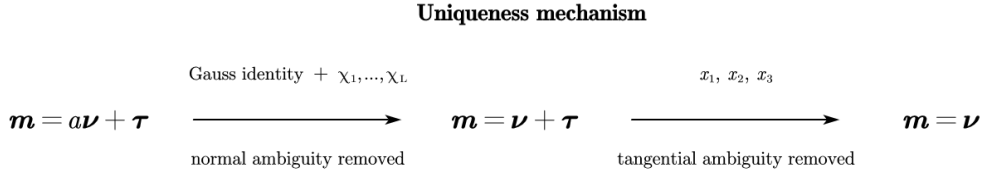


Fig. 2. Proof architecture. The decomposition $\mathbf{m} = a\boldsymbol{\nu} + \boldsymbol{\tau}$ separates normal-coefficient and tangential ambiguities. Topology-aware probes remove the former; coordinate probes remove the latter.

The unknown is an arbitrary smooth vector field

$$\mathbf{m} \in C^\infty(\Sigma; \mathbb{R}^3).$$

No unit-length or pointwise normality constraint is imposed. Relative to the true normal,

$$\mathbf{m} = a\boldsymbol{\nu} + \boldsymbol{\tau}, \quad a = \mathbf{m} \cdot \boldsymbol{\nu}, \quad \boldsymbol{\tau} \cdot \boldsymbol{\nu} = 0. \quad (8)$$

The first constraint is the boundary Gauss identity

$$\text{p. v.} \int_{\Sigma} \nabla_y G(x, y) \cdot \mathbf{m}(y) dS_y = -\frac{1}{2}, \quad x \in \Sigma. \quad (9)$$

Our sign convention is $G = 1/(4\pi|x - y|)$ with outward $\boldsymbol{\nu}$. The right-hand side $-1/2$ therefore selects the outward orientation.

For real-valued smooth densities ρ, η , define

$$B_m(\rho, \eta) = \int_{\Sigma} [(V\rho) \mathbf{m} \cdot \nabla^{\text{p.v.}} V\eta - (V\eta) \mathbf{m} \cdot \nabla^{\text{p.v.}} V\rho] dS. \quad (10)$$

We call

$$B_m(\rho, \eta) = 0 \quad \forall \rho, \eta \in C^\infty(\Sigma) \quad (11)$$

the *full weak Green-reciprocity condition*. It is the rigorous weak form of the antisymmetric boundary relation obtained by applying Green's second identity to harmonic potentials.

4 Identifiability of Full Green Reciprocity

THEOREM 4.1 (GENERAL UNIQUENESS THEOREM). *Let $\Omega \subset \mathbb{R}^3$ have bounded smooth finite topology. Then the outward unit normal $\boldsymbol{\nu}$ satisfies (9) and (11). Conversely, if $\mathbf{m} \in C^\infty(\Sigma; \mathbb{R}^3)$ satisfies both constraints, then*

$$\mathbf{m} = \boldsymbol{\nu} \quad \text{on } \Sigma.$$

Thus Gauss plus full weak Green reciprocity uniquely determines the outward unit normal for arbitrary bounded smooth finite topology.

The theorem is intentionally topology-independent: disconnected objects, internal cavities, and finite nesting do not produce an intrinsic ambiguity for the full formulation. Topology enters only when we ask how aggressively the full test family can be compressed.

4.1 Why the true normal satisfies every reciprocity probe

Before proving uniqueness, we verify feasibility in a form stronger than any finite-probe theorem requires.

PROPOSITION 4.2 (UNIVERSAL RECIPROCITY OF THE OUTWARD NORMAL). *For every pair of smooth densities ρ, η ,*

$$B_v(\rho, \eta) = 0.$$

Consequently the true outward normal satisfies the full reciprocity condition and every finite probe system obtained from it.

PROOF. Set

$$u = V\rho, \quad v = V\eta.$$

The boundary-gradient decomposition gives

$$\nabla^{\mathbf{p} \cdot \mathbf{v}} V\rho = \nabla_\Sigma u + \nu Au, \quad \nabla^{\mathbf{p} \cdot \mathbf{v}} V\eta = \nabla_\Sigma v + \nu Av.$$

Because ν is orthogonal to every tangential surface gradient,

$$\nu \cdot \nabla^{\mathbf{p} \cdot \mathbf{v}} V\rho = Au, \quad \nu \cdot \nabla^{\mathbf{p} \cdot \mathbf{v}} V\eta = Av.$$

Therefore

$$B_v(\rho, \eta) = \int_\Sigma (uAv - vAu) dS.$$

Self-adjointness of A gives

$$\int_\Sigma uAv dS = \int_\Sigma vAu dS,$$

so the two terms cancel. Hence

$$B_v(\rho, \eta) = 0.$$

Together with the Gauss identity (5), this proves existence for the full system as well as for every reduced system considered below. $\square$

4.2 The master fixed-probe identity

Fix a smooth boundary function ϕ and set

$$\eta_\phi = V^{-1}\phi.$$

For arbitrary ρ , write $u = V\rho$. Using (1), the fixed-probe condition $B_m(\rho, \eta_\phi) = 0$ becomes

$$0 = \int_\Sigma [u(\tau \cdot \nabla_\Sigma \phi + aA\phi) - \phi(\tau \cdot \nabla_\Sigma u + aAu)] dS.$$

Because Σ is closed, surface integration by parts gives

$$-\int_\Sigma \phi \tau \cdot \nabla_\Sigma u dS = \int_\Sigma u (\tau \cdot \nabla_\Sigma \phi + \phi \operatorname{div}_\Sigma \tau) dS.$$

Self-adjointness of A similarly gives

$$-\int_\Sigma a\phi Au dS = -\int_\Sigma u A(a\phi) dS.$$

Hence the fixed-probe equation is equivalent to

$$\int_\Sigma u [2\tau \cdot \nabla_\Sigma \phi + \phi \operatorname{div}_\Sigma \tau + aA\phi - A(a\phi)] dS = 0.$$

Since V maps smooth densities bijectively onto smooth boundary functions, $u = V\rho$ is arbitrary. Therefore

$$2\tau \cdot \nabla_\Sigma \phi + \phi \operatorname{div}_\Sigma \tau + aA\phi - A(a\phi) = 0. \tag{12}$$

This fixed-probe identity is the key ingredient for every finite-probe result.

The Gauss equation has an equally useful decomposed form. Since τ is tangential,

$$\nabla_y G(x, y) \cdot \tau(y) = \nabla_{\Sigma, y} G(x, y) \cdot \tau(y),$$

and surface integration by parts yields

$$\int_{\Sigma} \nabla_{\Sigma, y} G(x, y) \cdot \tau(y) dS_y = - \int_{\Sigma} G(x, y) \operatorname{div}_{\Sigma} \tau(y) dS_y.$$

Thus Gauss is equivalent to

$$Ka - V(\operatorname{div}_{\Sigma} \tau) = -\frac{1}{2}. \quad (13)$$

Now write

$$m = (1 + b)v + \tau, \quad b = a - 1.$$

The true normal satisfies $K1 = -1/2$. Subtracting its Gauss equation from (13) gives

$$Kb - V(\operatorname{div}_{\Sigma} \tau) = 0.$$

Applying V^{-1} and using $A = V^{-1}K$ yields

$$\operatorname{div}_{\Sigma} \tau = Ab. \quad (14)$$

For a threshold probe $\phi = \chi_k$ defined in (16), we have $\nabla_{\Sigma} \chi_k = 0$, so (12) and (14) give

$$\chi_k Ab + bA\chi_k - A(b\chi_k) = 0. \quad (15)$$

The identities above reduce uniqueness to controlling the scalar coefficient a and the tangential field τ . The topology-dependent mechanism is developed next.

5 Finite Reciprocity and Topology

5.1 Region tree and topology-aware probes

Consider the connected components of Ω and of $\mathbb{R}^3 \setminus \overline{\Omega}$. Build a region-adjacency graph $\mathcal{T}$ with one vertex for each region and one edge for each connected boundary component. Each closed embedded boundary component separates $\mathbb{R}^3$, so for finite topology $\mathcal{T}$ is a finite tree. Root it at the unbounded complementary region D_0 .

LEMMA 5.1 (FINITE REGION TREE). *For bounded smooth finite topology, the region-adjacency graph $\mathcal{T}$ is a finite tree rooted at the unique unbounded complementary region D_0 . In particular, the maximum boundary nesting depth L is finite.*

PROOF. Finiteness is immediate because Σ has finitely many connected components. To prove connectedness, choose points in any two regions and a smooth path joining them. After an arbitrarily small perturbation, the path is transverse to Σ and meets it only finitely many times. The successive regions visited by the path give a path in $\mathcal{T}$.

Now fix a connected boundary component Γ . By the Jordan–Brouwer separation theorem, $\mathbb{R}^3 \setminus \Gamma$ has exactly two connected components. The two regions adjacent to Γ lie on opposite sides of this separation. Every other boundary component is disjoint from Γ and, being connected, lies entirely on one of the two sides. Consequently a region-adjacency path cannot pass from one side of Γ to the other without traversing the edge corresponding to Γ . Thus every edge of $\mathcal{T}$ is a bridge. A finite connected graph in which every edge is a bridge is a tree.

Because Ω is bounded, $\mathbb{R}^3 \setminus \overline{\Omega}$ has a unique unbounded component D_0 , which supplies the root. Since the tree has finitely many edges, its maximum edge depth is finite. $\square$

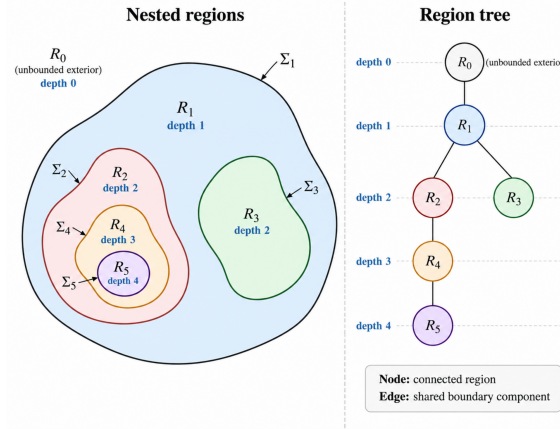


Fig. 3. Region-adjacency tree and nesting depth. The left panel shows a representative nested configuration. Each connected region becomes a vertex of the rooted tree on the right, and each connected boundary component becomes an edge joining its two adjacent regions. The unbounded complementary region D_0 is the root. Boundary depth is the depth of the corresponding edge, so branching increases the number of components at a fixed depth but does not increase the maximum nesting depth L .

For a boundary component Γ , let $d(\Gamma)$ be the depth of its edge: Γ separates a parent region at depth $k-1$ from a child at depth k . Crossing a boundary switches between Ω and its complement. Since the root D_0 lies in the complement, regions at odd vertex depth lie in Ω , whereas regions at even vertex depth lie in $\mathbb{R}^3 \setminus \overline{\Omega}$. Let

$$\Gamma^{(k)} = \bigcup_{d(\Gamma)=k} \Gamma, \quad L = \max_{\Gamma} d(\Gamma).$$

Define the threshold functions

$$\chi_1 \equiv 1, \quad \chi_k|_{\Gamma} = \begin{cases} 0, & d(\Gamma) < k, \\ 1, & d(\Gamma) \geq k, \end{cases} \quad k = 2, \dots, L, \quad (16)$$

and the corresponding densities

$$\zeta_k = V^{-1} \chi_k.$$

Because each χ_k is constant on every connected component of Σ , it is smooth on the disconnected manifold Σ .

The three geometric probes are the coordinate functions

$$\phi_j(x) = x_j, \quad \eta_j = V^{-1} \phi_j, \quad j = 1, 2, 3.$$

THEOREM 5.2 (FINITE COMPRESSION: $L+3$ PROBES SUFFICE). *Let L be the maximum boundary nesting depth. Suppose m satisfies the Gauss constraint (9) and*

$$B_m(\rho, \zeta_k) = 0 \quad \forall \rho, \quad k = 1, \dots, L,$$

together with

$$B_m(\rho, \eta_j) = 0 \quad \forall \rho, \quad j = 1, 2, 3.$$

Then $m = \nu$ on Σ .

THEOREM 5.3 (CONNECTED EXTERIOR: FOUR PROBES). *Assume that $\mathbb{R}^3 \setminus \overline{\Omega}$ is connected. Then the Gauss constraint together with the four fixed boundary probes*

$$1, \quad x_1, \quad x_2, \quad x_3,$$

with associated reciprocity densities obtained by applying V^{-1} , uniquely determines the outward unit normal. Equivalently, this is the depth-one case $L = 1$ of Theorem 5.2.

THEOREM 5.4 (CONNECTED DOMAIN WITH CAVITIES: FIVE PROBES). *Assume that Ω is connected and*

$$\mathbb{R}^3 \setminus \overline{\Omega} = D_0 \sqcup D_1 \sqcup \cdots \sqcup D_M, \quad M \geq 1,$$

where D_0 is unbounded and $D_1, \dots, D_M$ are bounded cavities. Define

$$\Gamma_0 = \partial D_0, \quad \Gamma_{\text{cav}} = \Sigma \setminus \Gamma_0,$$

and

$$\chi_{\text{cav}} = \begin{cases} 0, & \Gamma_0, \\ 1, & \Gamma_{\text{cav}}. \end{cases}$$

Then the five boundary probes

$$1, \quad \chi_{\text{cav}}, \quad x_1, \quad x_2, \quad x_3,$$

with associated reciprocity densities obtained by applying V^{-1} , together with Gauss uniquely determine the outward unit normal, independently of the number of cavities. This is the depth-two case $L = 2$.

The three statements above form one hierarchy rather than independent results. Theorems 5.3 and 5.4 are useful because they show the mechanism in familiar geometries, while Theorem 5.2 handles arbitrary finite nesting. The General Uniqueness Theorem then follows because full reciprocity contains every finite family selected by the compression theorem. In this sense, the finite-probe results quantify the information content of the full formulation rather than weaken its topological scope.

The formula $L + 3$ has a simple interpretation:

$$\underbrace{L}_{\text{topological layers}} + \underbrace{3}_{\text{ambient coordinate directions}}.$$

The first term removes ambiguity in the normal coefficient, the second removes the tangential field.

This decomposition of the probe count is structural. The number L does not count boundary components: a single χ_k acts simultaneously on every branch at depth k . Hence adding ten cavities at the same depth does not change the number of required topology probes, whereas inserting one object inside a cavity increases the maximum depth and therefore adds one threshold probe. The remaining three probes arise from the ambient coordinate directions and are independent of topology. This separation between topological depth and ambient dimension is the main compression principle behind the reduced systems.

5.2 A strict sign on each topological layer

The sign property follows from a simple but important Dirichlet structure of the threshold potentials.

LEMMA 5.5 (THRESHOLD-POTENTIAL STRUCTURE). *Fix $k \in \{2, \dots, L\}$ and let*

$$W_k = S(V^{-1}\chi_k).$$

Let $\Gamma \subset \Gamma^{(k)}$ separate its parent region R_{k-1} from its child region R_k in the rooted region tree. Then

$$W_k \equiv 1 \quad \text{in } R_k, \quad (17)$$

whereas

$$0 < W_k < 1 \quad \text{in } R_{k-1}. \quad (18)$$

The same statement holds simultaneously for every branch at depth k .

PROOF. The single-layer potential is continuous across Σ and its trace is χ_k . By definition of the threshold function, every boundary component of the child region R_k has depth at least k . Hence the boundary value of W_k is one on all of ∂R_k . Since W_k is harmonic in R_k , uniqueness of the Dirichlet problem gives

$$W_k \equiv 1.$$

The parent region R_{k-1} has one parent-facing boundary component at depth $k-1$, on which $\chi_k = 0$, and at least one child-facing component at depth k , on which $\chi_k = 1$. Other child-facing components, if present, also carry value one because their depths are at least k . Thus the boundary data of W_k on R_{k-1} take values in $\{0, 1\}$ and are not constant. The weak maximum principle gives

$$0 \leq W_k \leq 1 \quad \text{in } R_{k-1},$$

and the strong maximum principle improves this to the strict inequalities (18). Nothing in the argument depends on how many child branches emanate from R_{k-1} . The single threshold trace treats them all simultaneously. $\square$

COROLLARY 5.6 (BRANCHING DOES NOT INCREASE THE PROBE COUNT). *At any fixed depth k , one threshold probe χ_k is sufficient to expose all boundary components in $\Gamma^{(k)}$, regardless of their number.*

PROOF. For every $\Gamma \subset \Gamma^{(k)}$, Lemma 5.5 produces the same strict parent/child contrast for the single potential W_k . The Hopf-sign argument below can therefore be applied on every such Γ using the same probe. $\square$

PROPOSITION 5.7 (THRESHOLD-PROBE SIGN). *For each $k = 1, \dots, L$,*

$$(-1)^k A\chi_k > 0 \quad \text{on } \Gamma^{(k)}. \quad (19)$$

PROOF. Let

$$W_k(z) = S(V^{-1}\chi_k)(z).$$

The single-layer potential is harmonic in every complementary and material region, is continuous across Σ , and has trace χ_k .

For $k = 1$, the trace is identically one. Hence $W_1 \equiv 1$ in every bounded region. In the unbounded exterior D_0 , W_1 is harmonic, equals one on the boundary, and tends to zero at infinity. Therefore $0 < W_1 < 1$ in D_0 . If n_0 denotes the outward normal of D_0 , then $n_0 = -\nu$ on $\Gamma^{(1)}$, and Hopf's boundary lemma gives $\partial_{n_0} W_1 > 0$. Equivalently, the exterior-side derivative in the ν direction is strictly negative. The interior derivative is zero because the bounded child region carries the constant solution one. The principal-value normal trace is the average of the two one-sided traces, so

$$A\chi_1 = A1 < 0 \quad \text{on } \Gamma^{(1)}.$$

Now let $k \geq 2$ and fix

$$\Gamma \subset \Gamma^{(k)}.$$

Let R_{k-1} and R_k be the adjacent parent and child regions, and let n_{k-1} be the outward unit normal of the parent region along Γ . By Lemma 5.5,

$$0 < W_k < 1 \quad \text{in } R_{k-1}, \quad W_k \equiv 1 \quad \text{in } R_k.$$

Thus W_k attains a strict boundary maximum on Γ when approached from the parent side. Hopf's boundary lemma gives

$$\partial_{n_{k-1}} W_k > 0 \quad \text{on } \Gamma,$$

whereas the normal derivative from the child side is zero because W_k is constant there.

The parity statement following Lemma 5.1 gives

$$n_{k-1} = (-1)^k \nu \quad \text{on } \Gamma.$$

Hence the parent-side derivative in the ν direction has sign $(-1)^k$, while the child-side derivative vanishes. Since $A\chi_k$ is the arithmetic mean of the two one-sided ν -derivatives,

$$(-1)^k A\chi_k > 0 \quad \text{on } \Gamma.$$

Because $\Gamma \subset \Gamma^{(k)}$ was arbitrary, the inequality holds on the whole k th layer. $\square$

The significance is geometric rather than spectral: one threshold function acts simultaneously on every branch at depth k . Branching therefore does not increase the probe count.

5.3 Layer-by-layer elimination of the normal ambiguity

We prove Theorem 5.2 by induction over the boundary depth.

Base layer. For $k = 1$, $\chi_1 \equiv 1$, hence

$$b\chi_1 = b.$$

Equation (15) reduces to

$$bA\chi_1 = 0.$$

By Proposition 5.7,

$$A\chi_1 < 0 \quad \text{on } \Gamma^{(1)},$$

so

$$b = 0 \quad \text{on } \Gamma^{(1)}.$$

Induction step. Assume

$$b = 0 \quad \text{on } \Gamma^{(1)} \cup \dots \cup \Gamma^{(k-1)}.$$

By construction, $\chi_k = 0$ on all layers shallower than k and $\chi_k = 1$ on every layer of depth at least k . The induction hypothesis therefore implies

$$b\chi_k = b \quad \text{on all of } \Sigma.$$

Equation (15) becomes

$$(\chi_k - 1)Ab + bA\chi_k = 0.$$

Restricting to $\Gamma^{(k)}$, where $\chi_k = 1$, gives

$$bA\chi_k = 0.$$

The strict sign in Proposition 5.7 shows that $A\chi_k$ is nowhere zero on the k th layer. Hence

$$b = 0 \quad \text{on } \Gamma^{(k)}.$$

Proceeding through $k = L$ exhausts every connected boundary component and gives

$$a = 1 \quad \text{on all of } \Sigma. \quad (20)$$

Equation (14) then gives

$$\operatorname{div}_\Sigma \tau = 0.$$

Use the three coordinate probes in (12). Since $a = 1$ and $\operatorname{div}_\Sigma \tau = 0$,

$$\tau \cdot \nabla_\Sigma x_j = 0, \quad j = 1, 2, 3.$$

But

$$\nabla_\Sigma x_j = e_j - (e_j \cdot \nu)\nu$$

and $\tau \cdot \nu = 0$, so these equations are precisely $\tau \cdot e_j = 0$ for $j = 1, 2, 3$. Thus

$$\tau = 0,$$

and $m = \nu$. This proves Theorem 5.2.

PROOFS OF THEOREMS 5.3 AND 5.4. If $\mathbb{R}^3 \setminus \overline{\Omega}$ is connected, every boundary component is adjacent to the unbounded complementary region, so every edge has depth one. Hence $L = 1$ and the only threshold function is $\chi_1 \equiv 1$. Theorem 5.2 therefore gives exactly the four boundary probes $1, x_1, x_2, x_3$.

If Ω is connected and has at least one bounded cavity, the rooted region tree has the unbounded complement at depth zero, the single material region Ω at depth one, and every cavity at depth two. Thus $L = 2$, and the second threshold function is precisely $\chi_2 = \chi_{\text{cav}}$. Theorem 5.2 therefore gives the stated five-probe system. $\square$

Finally, bounded smooth finite topology has only finitely many boundary components, hence finite maximum depth L . Full reciprocity (11) contains, in particular, all L threshold families and the three coordinate families. Theorem 5.2 together with Proposition 4.2 therefore prove Theorem 4.1.

6 Well-Posedness under Perturbed Data

The unperturbed theory determines the kernel of the inverse problem. We now characterize its range and the continuity of the inverse on that range. This separates exact realizability, uniqueness, and conditioning of perturbed data.

Define

$$\mathcal{G}m := \text{p.v.} \int_\Sigma \nabla_y G(\cdot, y) \cdot m(y) dS_y.$$

Consider perturbed data

$$\mathcal{G}m^\delta = -\frac{1}{2} + g^\delta, \quad (21)$$

and

$$B_{m^\delta}(\rho, \eta) = E^\delta(\rho, \eta) \quad \forall \rho, \eta \in C^\infty(\Sigma). \quad (22)$$

We call E^δ *regular* if there is a linear operator

$$\mathcal{R}^\delta : C^\infty(\Sigma) \rightarrow C^\infty(\Sigma)$$

such that

$$E^\delta(\rho, \eta) = \int_{\Sigma} (V\rho) \mathcal{R}^\delta(V\eta) dS \quad \forall \rho, \eta \in C^\infty(\Sigma). \quad (23)$$

For data generated by a smooth exact solution, this representation is automatic. Define

$$\gamma^\delta := V^{-1}g^\delta. \quad (24)$$

For a scalar function b and a tangential field τ , set

$$\mathcal{D}_{b,\tau} := 2\tau \cdot \nabla_{\Sigma} + M_{\text{div}_{\Sigma} \tau} + [M_b, A], \quad (25)$$

where M_f denotes multiplication by f and $[M_b, A] = M_b A - A M_b$.

THEOREM 6.1 (EXACT SOLVABILITY AND RANGE CHARACTERIZATION). *Assume $g^\delta \in C^\infty(\Sigma)$ and that E^δ is regular with representing operator $\mathcal{R}^\delta$. Then (21)–(22) has a smooth solution if and only if there exist*

$$b \in C^\infty(\Sigma), \quad \tau \in C^\infty(T\Sigma),$$

such that

$$\gamma^\delta = Ab - \text{div}_{\Sigma} \tau, \quad (26)$$

and

$$\mathcal{R}^\delta = \mathcal{D}_{b,\tau} = 2\tau \cdot \nabla_{\Sigma} + M_{\text{div}_{\Sigma} \tau} + [M_b, A]. \quad (27)$$

When these conditions hold, the solution is

$$m^\delta = (1 + b)v + \tau, \quad (28)$$

and it is unique.

PROOF OF THEOREM 6.1. Suppose first that m^δ is a smooth exact solution and write

$$m^\delta - v = bv + \tau, \quad \tau \cdot v = 0.$$

Subtracting the Gauss equation of v from (21) gives

$$Kb - V(\text{div}_{\Sigma} \tau) = g^\delta.$$

Applying V^{-1} yields (26).

Let

$$u = V\rho, \quad v = V\eta.$$

Since $B_v(\rho, \eta) = 0$,

$$E^\delta(\rho, \eta) = B_{bv+\tau}(\rho, \eta).$$

Using

$$\nabla^{\text{p.v.}} V\rho = \nabla_{\Sigma} u + vAu, \quad \nabla^{\text{p.v.}} V\eta = \nabla_{\Sigma} v + vAv,$$

we obtain

$$E^\delta(\rho, \eta) = \int_{\Sigma} [u \tau \cdot \nabla_{\Sigma} v - v \tau \cdot \nabla_{\Sigma} u + b(uAv - vAu)] dS.$$

Because Σ is closed,

$$-\int_{\Sigma} v \tau \cdot \nabla_{\Sigma} u dS = \int_{\Sigma} u (\tau \cdot \nabla_{\Sigma} v + v \text{div}_{\Sigma} \tau) dS.$$

Self-adjointness of A gives

$$-\int_{\Sigma} b v A u dS = -\int_{\Sigma} u A(bv) dS.$$

Therefore

$$E^{\delta}(\rho, \eta) = \int_{\Sigma} u [2\tau \cdot \nabla_{\Sigma} v + (\operatorname{div}_{\Sigma} \tau) v + bAv - A(bv)] dS.$$

By the uniqueness of the representing operator in (23),

$$\mathcal{R}^{\delta} = 2\tau \cdot \nabla_{\Sigma} + M_{\operatorname{div}_{\Sigma} \tau} + [M_b, A],$$

which is (27).

Conversely, suppose b and tangential τ satisfy (26)–(27), and define

$$m^{\delta} = (1 + b)v + \tau.$$

Reversing the preceding calculation gives

$$B_{m^{\delta}}(\rho, \eta) = E^{\delta}(\rho, \eta)$$

for every ρ, η , while (26) is equivalent to the perturbed Gauss equation. Thus m^{δ} is an exact solution.

If m_1^{δ} and m_2^{δ} are two exact solutions for the same data, their difference h satisfies

$$\mathcal{G}h = 0, \quad B_h(\rho, \eta) = 0 \quad \forall \rho, \eta.$$

Hence $v + h$ satisfies the unperturbed full system. By Theorem 4.1, $h = 0$. □

Equivalently, define the linear data operator

$$\mathfrak{L}(b, \tau) := (Ab - \operatorname{div}_{\Sigma} \tau, \mathcal{D}_{b, \tau}). \quad (29)$$

Then smooth exact solvability is equivalent to

$$(\gamma^{\delta}, \mathcal{R}^{\delta}) \in \operatorname{Ran} \mathfrak{L},$$

while the General Uniqueness Theorem implies

$$\ker \mathfrak{L} = \{0\}.$$

Here and below, T^* denotes the formal $L^2(\Sigma)$ adjoint on smooth functions.

COROLLARY 6.2 (NECESSARY COMPATIBILITY CONDITIONS). *If the perturbed system is exactly solvable, then*

$$(\mathcal{R}^{\delta})^* = -\mathcal{R}^{\delta}, \quad (30)$$

equivalently

$$E^{\delta}(\rho, \eta) = -E^{\delta}(\eta, \rho), \quad E^{\delta}(\rho, \rho) = 0. \quad (31)$$

Moreover, $\mathcal{R}^{\delta}$ has the structure

$$\mathcal{R}^{\delta} = \underbrace{2\tau \cdot \nabla_{\Sigma} + M_{\operatorname{div}_{\Sigma} \tau}}_{\text{first-order skew transport}} + \underbrace{[M_b, A]}_{\text{order at most } -1}. \quad (32)$$

Hence arbitrarily small perturbations of the right-hand side need not be exactly solvable.

PROOF OF COROLLARY 6.2. For tangential τ , the formal adjoint of the first-order operator $\tau \cdot \nabla_{\Sigma}$ on a closed surface is

$$(\tau \cdot \nabla_{\Sigma})^* = -\tau \cdot \nabla_{\Sigma} - M_{\operatorname{div}_{\Sigma} \tau}.$$

Hence

$$(2\tau \cdot \nabla_\Sigma + M_{\text{div}_\Sigma \tau})^* = -(2\tau \cdot \nabla_\Sigma + M_{\text{div}_\Sigma \tau}).$$

Since $A = A^*$ and $M_b = M_b^*$,

$$[M_b, A]^* = -[M_b, A].$$

Equation (27) therefore implies

$$(\mathcal{R}^\delta)^* = -\mathcal{R}^\delta,$$

which is equivalent to (31).

Moreover, A is a classical pseudodifferential operator of order zero [Ando et al. 2025], so $[M_b, A]$ has order at most -1 . This proves (32). In particular, a nonzero symmetric bilinear perturbation, however small, cannot be represented by an exactly solvable system. $\square$

The range characterization can be tested constructively from the same $L + 3$ slices used in the uniqueness theorem. Let

$$\Phi_L = \{\chi_1, \dots, \chi_L, x_1, x_2, x_3\}, \quad r_\phi^\delta := \mathcal{R}^\delta \phi.$$

THEOREM 6.3 (FINITE-SLICE SOLVABILITY TEST). *Given regular data (g^δ, E^δ) , define a scalar field $b^\#$ recursively over the boundary layers. Set $b_{<1}^\# = 0$. Once $b^\#$ is known on all layers of depth $< k$, let $b_{<k}^\#$ equal this known function on those layers and zero on all deeper layers, and define on $\Gamma^{(k)}$*

$$b^\# = \frac{r_{\chi_k}^\delta + \gamma^\delta - Ab_{<k}^\#}{A\chi_k}. \quad (33)$$

After all layers are determined, define an ambient vector field $\tau^\#$ componentwise by

$$2\tau_j^\# = r_{x_j}^\delta + x_j \gamma^\delta - x_j Ab^\# - b^\# Ax_j + A(b^\# x_j), \quad j = 1, 2, 3. \quad (34)$$

Then the full perturbed system is exactly solvable if and only if

$$\tau^\# \cdot \nu = 0, \quad (35)$$

$$\gamma^\delta = Ab^\# - \text{div}_\Sigma \tau^\#, \quad (36)$$

$$\mathcal{R}^\delta = 2\tau^\# \cdot \nabla_\Sigma + M_{\text{div}_\Sigma \tau^\#} + [M_{b^\#}, A]. \quad (37)$$

In that case

$$m^\delta = (1 + b^\#)\nu + \tau^\#$$

is the unique exact solution.

PROOF OF THEOREM 6.3. Assume first that the data are exactly solvable, and let

$$m^\delta - \nu = b\nu + \tau.$$

By Theorem 6.1,

$$\gamma^\delta = Ab - \text{div}_\Sigma \tau, \quad r_\phi^\delta = \mathcal{R}^\delta \phi = 2\tau \cdot \nabla_\Sigma \phi + \phi \text{div}_\Sigma \tau + bA\phi - A(b\phi).$$

Eliminating $\text{div}_\Sigma \tau$ gives

$$2\tau \cdot \nabla_\Sigma \phi + \phi Ab + bA\phi - A(b\phi) = r_\phi^\delta + \phi \gamma^\delta. \quad (38)$$

For $\phi = \chi_k$, $\nabla_\Sigma \chi_k = 0$. If $b_{<k}$ denotes b on layers of depth $< k$ and zero elsewhere, then

$$b\chi_k = b - b_{<k}.$$

Restricting (38) to $\Gamma^{(k)}$ gives

$$bA\chi_k = r_{\chi_k}^\delta + \gamma^\delta - Ab_{<k}.$$

Proposition 5.7 shows that $A\chi_k$ never vanishes on $\Gamma^{(k)}$. Induction over k therefore proves

$$b^\# = b.$$

For $\phi = x_j$,

$$\tau \cdot \nabla_\Sigma x_j = \tau_j,$$

and (38) gives exactly (34). Hence

$$\tau^\# = \tau.$$

The three conditions (35)–(37) are therefore necessary.

Conversely, suppose the recursively constructed pair $(b^\#, \tau^\#)$ satisfies (35)–(37). Condition (35) makes $\tau^\#$ tangential, and the other two conditions are precisely

$$\gamma^\delta = Ab^\# - \operatorname{div}_\Sigma \tau^\#, \quad \mathcal{R}^\delta = \mathcal{D}_{b^\#, \tau^\#}.$$

Theorem 6.1 then implies exact solvability and gives the stated unique solution. $\square$

The first $L + 3$ slices therefore reconstruct the only possible candidate solution; the remaining full reciprocity data impose the global consistency condition (37).

THEOREM 6.4 (CONDITIONAL LIPSCHITZ STABILITY ON THE ADMISSIBLE RANGE). *Let Ω have bounded smooth finite topology and maximum boundary nesting depth L . Fix $s \geq 0$. Suppose the perturbed data are exactly solvable and regular, with*

$$\gamma^\delta \in H^s(\Sigma), \quad r_\phi^\delta \in H^s(\Sigma) \quad (\phi \in \Phi_L).$$

Then there is a constant $C_{\Omega, s} > 0$ such that

$$\|m^\delta - v\|_{H^s(\Sigma)} \leq C_{\Omega, s} \left(\|\gamma^\delta\|_{H^s(\Sigma)} + \sum_{k=1}^L \|r_{\chi_k}^\delta\|_{H^s(\Sigma)} + \sum_{j=1}^3 \|r_{x_j}^\delta\|_{H^s(\Sigma)} \right). \quad (39)$$

Since V^{-1} is an elliptic pseudodifferential operator of order 1,

$$\|m^\delta - v\|_{H^s(\Sigma)} \leq C'_{\Omega, s} \left(\|g^\delta\|_{H^{s+1}(\Sigma)} + \sum_{k=1}^L \|r_{\chi_k}^\delta\|_{H^s(\Sigma)} + \sum_{j=1}^3 \|r_{x_j}^\delta\|_{H^s(\Sigma)} \right). \quad (40)$$

Thus the inverse of $\mathfrak{Q}$ is Lipschitz on its smooth range when the data are measured through the transformed Gauss residual and the same $L + 3$ reciprocity slices.

PROOF OF THEOREM 6.4. Let

$$m^\delta - v = bv + \tau.$$

By Theorem 6.1,

$$\operatorname{div}_\Sigma \tau = Ab - \gamma^\delta,$$

and for every $\phi \in \Phi_L$,

$$2\tau \cdot \nabla_\Sigma \phi + \phi Ab + bA\phi - A(b\phi) = r_\phi^\delta + \phi \gamma^\delta. \quad (41)$$

For $\phi = \chi_k$, let $b_{<k} = (1 - \chi_k)b$. On $\Gamma^{(k)}$,

$$bA\chi_k = r_{\chi_k}^\delta + \gamma^\delta - Ab_{<k}. \quad (42)$$

By Proposition 5.7,

$$c_k = \min_{\Gamma^{(k)}} |A\chi_k| > 0.$$

Multiplication by $(A\chi_k)^{-1}$ is bounded on $H^s(\Gamma^{(k)})$, and A is bounded on $H^s(\Sigma)$. Thus

$$\|b\|_{H^s(\Gamma^{(k)})} \leq C_k \left(\|r_{\chi_k}^\delta\|_{H^s} + \|\gamma^\delta\|_{H^s} + \|b_{<k}\|_{H^s} \right).$$

Induction over k yields

$$\|b\|_{H^s(\Sigma)} \leq C_{\Omega,s}^{(1)} \left(\|\gamma^\delta\|_{H^s} + \sum_{k=1}^L \|r_{\chi_k}^\delta\|_{H^s} \right). \quad (43)$$

For $\phi = x_j$, (41) gives

$$2\tau_j = r_{x_j}^\delta + x_j \gamma^\delta - x_j Ab - bAx_j + A(bx_j).$$

Smooth multiplication and the order-zero boundedness of A imply

$$\|\tau\|_{H^s} \leq C_{\Omega,s}^{(2)} \left(\sum_{j=1}^3 \|r_{x_j}^\delta\|_{H^s} + \|\gamma^\delta\|_{H^s} + \|b\|_{H^s} \right).$$

Combining this with (43) and $m^\delta - v = bv + \tau$ proves (39). Finally,

$$\|V^{-1}g^\delta\|_{H^s} \leq C\|g^\delta\|_{H^{s+1}}$$

gives (40). □

REMARK 6.5 (GEOMETRY DEPENDENCE OF THE STABILITY CONSTANT). *For each depth k ,*

$$c_k := \min_{y \in \Gamma^{(k)}} |A\chi_k(y)| > 0.$$

The constant $C_{\Omega,s}$ depends on the fixed geometry, the operator norm of A , the depth L , and the quantities c_k^{-1} . No geometry-uniform bound is asserted for degenerating families in which some $c_k \rightarrow 0$.

7 Analytic Case Studies

These exact case studies expose the mechanisms of the theory in closed form.

7.1 Sphere

Let $\Omega = B_R(0)$. With our normalization of G , the single-layer potential of constant density $1/R$ equals one on the sphere, so

$$q = V^{-1}1 = \frac{1}{R}, \quad A1 = -\frac{1}{2R}.$$

Thus the constant probe has nonzero sign everywhere. Gauss identity plus this one probe fixes $a = 1$, and the three coordinate probes eliminate τ . This is the depth-one four-probe case.

7.2 Spherical shell

Let

$$\Omega = \{x : r < |x| < R\}.$$

The equilibrium potential with boundary value one is generated entirely by the outer sphere:

$$q = \begin{cases} 1/R, & |x| = R, \\ 0, & |x| = r. \end{cases}$$

The vanishing on the inner sphere shows exactly why the constant probe cannot see the cavity coefficient. The cavity-separating function χ_{cav} is zero on $|x| = R$ and one on $|x| = r$. Its harmonic extension is strictly between zero and one in the shell and constant one in the cavity. The Hopf lemma therefore yields $A\chi_{\text{cav}} > 0$ on the inner sphere, closing the blind spot.

7.3 Three-level nesting

Consider the concentric configuration

$$\Omega = \{R_2 < |x| < R_1\} \cup \{|x| < R_3\}, \quad R_3 < R_2 < R_1.$$

The three boundary spheres have depths 1, 2, 3. The threshold traces are

$$\chi_1 = (1, 1, 1), \quad \chi_2 = (0, 1, 1), \quad \chi_3 = (0, 0, 1),$$

listed from outermost to innermost sphere. They determine the normal coefficient layer by layer. The coordinate probes then remove the tangential field. The theorem therefore uses $3 + 3 = 6$ probes. Adding more branches at any of these depths would not increase the count.

8 Relation to Existing Green–Gauss Formulations

G3R [Wu et al. 2026] uses Green’s third identity for harmonic functions. Let U be harmonic in Ω , let $\phi = U|_{\Sigma}$, and define

$$(\mathcal{G}[U]m)(x) = \int_{\Sigma} [G(x, y)\nabla U(y) - U(y)\nabla_y G(x, y)] \cdot m(y) dS_y.$$

Its boundary equation is

$$\mathcal{G}[U]m = \frac{1}{2}\phi.$$

For $U \equiv 1$ this is the Gauss equation. The low-order family $\{1, x_1, x_2, x_3\}$ therefore appears naturally in both G3R and our reduced reciprocity systems.

The equations are nevertheless different before the normal coefficient is known. The exact relation is as follows.

PROPOSITION 8.1 (G3R RESIDUAL VERSUS RECIPROCITY RESIDUAL). *Let $\eta_\phi = V^{-1}\phi$, write $m = av + \tau$, and let $v = V\rho$. Then*

$$\begin{aligned} \int_{\Sigma} \rho(x) \left[(\mathcal{G}[U]m)(x) - \frac{1}{2}\phi(x) \right] dS_x \\ = B_m(\rho, \eta_\phi) + \frac{1}{2} \int_{\Sigma} v \eta_\phi (a - 1) dS. \end{aligned} \tag{44}$$

PROOF. Let

$$w = S\eta_\phi.$$

Since $V\eta_\phi = \phi = U|_\Sigma$, uniqueness of the interior Dirichlet problem gives

$$w = U \quad \text{in } \Omega.$$

The interior gradient trace of the single-layer potential is

$$\nabla U = \nabla^{p,v} V\eta_\phi + \frac{1}{2}\eta_\phi v.$$

Pair the G3R residual with ρ and interchange the order of integration:

$$\int_\Sigma \rho(x) \mathcal{G}[U] m(x) dS_x = \int_\Sigma [(V\rho) m \cdot \nabla U - U m \cdot \nabla^{p,v} V\rho] dS.$$

Write $v = V\rho$ and substitute the interior trace:

$$= \int_\Sigma [v m \cdot \nabla^{p,v} V\eta_\phi - \phi m \cdot \nabla^{p,v} V\rho] dS + \frac{1}{2} \int_\Sigma v \eta_\phi dS.$$

The first integral is $B_m(\rho, \eta_\phi)$. For the prescribed boundary term,

$$\frac{1}{2} \int_\Sigma \rho \phi dS = \frac{1}{2} \int_\Sigma \rho V\eta_\phi dS = \frac{1}{2} \int_\Sigma (V\rho) \eta_\phi dS = \frac{1}{2} \int_\Sigma v \eta_\phi dS,$$

where self-adjointness of V was used. Subtracting gives exactly (44). $\square$

9 Discussion and Limitations

The results concern the continuous inverse problem. We establish that the continuous target is uniquely determined and quantify a finite sufficient subset of the reciprocity information. The continuous perturbation theory above does provide an exact range characterization and a geometry-dependent Lipschitz conditioning estimate on the admissible data range.

The topology-aware probes should be interpreted similarly. The full Green-reciprocity formulation is topology-independent. It requires no prior knowledge of nesting depth. The $L + 3$ theorem is a structural compression result explaining how much of the infinite family is sufficient once the finite region tree is known.

The assumptions are also intentionally classical. We require a compact smooth embedded boundary with finitely many connected components. Open surfaces, touching components, self-intersections, non-manifold boundaries, lower regularity, or infinitely accumulating boundary components are outside the theorem. These cases would require additional trace and topology arguments.

Within this scope, the theory separates four logically distinct questions: identifiability, finite information requirement, exact solvability, and stability. The General Uniqueness Theorem resolves identifiability. The $L + 3$ theorem bounds the finite information requirement. The range theorem characterizes exact solvability and Theorem 6.4 gives continuous Lipschitz stability on the admissible range.

Discretization and numerical solution are deliberately left to future work. The results established here are properties of the continuous boundary integral system, and they do not imply that a particular discretization inherits the same identifiability, exact solvability, or stability properties. Geometry approximation, quadrature, the choice of discrete trial and test spaces, and numerical regularization may all alter the nullspace, range, and conditioning of the resulting finite-dimensional system. Designing structure-preserving discretizations for the continuous formulation, and establishing discrete analogues of the uniqueness, solvability, and stability results proved here, therefore remain challenging research

problems. We view both the development of reliable numerical solvers and the theoretical analysis of the corresponding discrete systems as important directions for future work.

10 Conclusion

We introduced Green reciprocity as a continuous global constraint for normal orientation and developed a unified well-posedness theory for the resulting inverse problem. The full Gauss–reciprocity system has the outward unit normal as its unique smooth solution on arbitrary bounded smooth finite topology. The proof does not assume normality or unit length of the unknown field.

Topology enters only when the infinite reciprocity family is compressed. The rooted region tree supplies boundary layers $\Gamma^{(1)}, \dots, \Gamma^{(L)}$, and the threshold functions $\chi_1, \dots, \chi_L$ have strict alternating Hopf signs on these layers. They determine the normal coefficient successively, while the three coordinate probes eliminate the tangential remainder. Thus $L + 3$ fixed probe families retain the full uniqueness guarantee.

For perturbed data, solvability is characterized exactly by the range of the linear map

$$(b, \tau) \mapsto (Ab - \operatorname{div}_\Sigma \tau, 2\tau \cdot \nabla_\Sigma + M_{\operatorname{div}_\Sigma \tau} + [M_b, A]).$$

The same $L + 3$ slices reconstruct the only possible pair (b, τ) . The remaining full reciprocity operator data are a consistency condition rather than additional freedom. Small perturbations need not be exactly solvable, but on the admissible range the inverse is unique and Lipschitz continuous.

A Layer-Potential Facts

We collect the analytic facts used in the main text and record the sign conventions explicitly.

LEMMA A.1 (LAYER-POTENTIAL IDENTITIES). *On a smooth closed boundary in $\mathbb{R}^3$,*

$$V : H^{-1/2}(\Sigma) \rightarrow H^{1/2}(\Sigma)$$

is a self-adjoint elliptic isomorphism, the Plemelj identity

$$KV = VK'$$

holds, and

$$A := V^{-1}K = K'V^{-1}$$

is $L^2(\Sigma)$ -self-adjoint and is a classical pseudodifferential operator of order zero. Moreover,

$$K1 = -\frac{1}{2},$$

and for every smooth density ρ ,

$$\nabla^{\mathbf{p} \cdot \mathbf{v}} V \rho = \nabla_\Sigma(V \rho) + v K' \rho.$$

PROOF. The mapping property, trace and jump relations, and Plemelj symmetrization are standard facts for Laplace layer potentials; see McLean [2000]. The order-zero pseudodifferential characterization of $A = V^{-1}K$ on smooth three-dimensional boundaries is recorded, for example, by Ando et al. [2025]. Since $V = V^*$ and $K' = K^*$,

$$A^* = (V^{-1}K)^* = K'V^{-1} = A.$$

For $K1$, let z lie in one component of Ω . Summing the divergence theorem over all connected components gives

$$\int_{\Sigma} \partial_{vy} G(z, y) dS_y = -1.$$

If $z \in \mathbb{R}^3 \setminus \overline{\Omega}$, whether in a bounded cavity or in the unbounded exterior, the same integral is zero. Thus across every boundary component the two limiting values are -1 on the Ω side and 0 on the complementary side. The principal-value boundary value is their average, hence

$$K1 = -\frac{1}{2}.$$

Finally, the tangential derivative of the continuous single-layer trace is $\nabla_{\Sigma}(V\rho)$, whereas the principal-value normal component is $K'\rho$. More precisely, the one-sided traces satisfy

$$\partial_v^i S\rho = K'\rho + \frac{1}{2}\rho, \quad \partial_v^e S\rho = K'\rho - \frac{1}{2}\rho.$$

Their average is $K'\rho$, while their difference is $-\rho$. Combining the tangential trace with this averaged normal trace gives

$$\nabla^{p.v.} V\rho = \nabla_{\Sigma}(V\rho) + \nu K'\rho.$$

This also explains why the principal-value operator A appearing in the main proof is the average normal response of the single-layer harmonic extension rather than either one-sided derivative separately. $\square$

LEMMA A.2 (EQUILIBRIUM DENSITY). *Let*

$$q = V^{-1}1.$$

If $\mathbb{R}^3 \setminus \overline{\Omega}$ is connected, then

$$q > 0 \quad \text{on } \Sigma,$$

and

$$A1 = -\frac{1}{2}q.$$

For a connected domain with bounded cavities,

$$q > 0 \quad \text{on the outer boundary,}$$

and

$$q = 0 \quad \text{on every cavity boundary.}$$

PROOF. Define

$$U = Sq.$$

Since $U|_{\Sigma} = Vq = 1$, the interior Dirichlet problem gives $U \equiv 1$ in every connected component of Ω .

If the exterior is connected, then U is harmonic there, equals one on the boundary, and decays to zero at infinity. The strong maximum principle gives $0 < U < 1$ in the exterior, and Hopf's lemma yields

$$\partial_v^e U < 0.$$

On the interior side,

$$\partial_v^i U = 0.$$

The single-layer jump relation

$$\partial_v^e U - \partial_v^i U = -q$$

therefore implies

$$q > 0.$$

Since $K1 = -1/2$,

$$A1 = V^{-1}K1 = -\frac{1}{2}q.$$

If bounded cavities are present, U has constant boundary value one on each cavity and hence $U \equiv 1$ inside that cavity as well. Thus both one-sided normal derivatives vanish there, and the jump relation gives $q = 0$ on the cavity boundary. The outer-boundary argument is unchanged and still gives $q > 0$ there. $\square$

B A Three-Probe Refinement at Depth One

The three coordinate functions are universal but not always minimal. The following refinement makes the geometric requirement precise.

COROLLARY B.1 (THREE-PROBE UNIQUENESS UNDER DENSE TANGENT SPANNING). *Assume the connected-exterior hypothesis of Theorem 5.3. Let*

$$\phi_0 = 1, \quad \phi_1, \phi_2 \in C^\infty(\Sigma), \quad \eta_i = V^{-1}\phi_i.$$

Suppose

$$\Sigma_* = \{y \in \Sigma : \nabla_\Sigma \phi_1(y), \nabla_\Sigma \phi_2(y) \text{ are linearly independent}\}$$

is dense in Σ . If m satisfies Gauss and the three reciprocity families associated with ϕ_0, ϕ_1, ϕ_2 , then

$$m = v.$$

PROOF. The constant probe and Gauss give

$$a = 1, \quad \operatorname{div}_\Sigma \tau = 0$$

by the depth-one step in the proof of Theorem 5.2. The master identity (12) applied to ϕ_1 and ϕ_2 therefore reduces to

$$\tau \cdot \nabla_\Sigma \phi_1 = 0, \quad \tau \cdot \nabla_\Sigma \phi_2 = 0.$$

On Σ_* the two gradients span the tangent plane, so $\tau = 0$ there. Since τ is continuous and Σ_* is dense,

$$\tau \equiv 0 \quad \text{on } \Sigma.$$

Hence $m = v$. $\square$

For example, one may take

$$\phi_1(y) = e_1 \cdot y, \quad \phi_2(y) = e_2 \cdot y,$$

provided

$$v(y) \cdot (e_1 \times e_2) \neq 0$$

on a dense subset of Σ .

The result is close to minimal after the normal coefficient has been fixed. One nonconstant probe ϕ gives only

$$\tau \cdot \nabla_\Sigma \phi = 0.$$

On an oriented surface, let J denote rotation by $\pi/2$ in the tangent plane. Then fields of the form

$$\tau = F(\phi) J \nabla_\Sigma \phi$$

are orthogonal to $\nabla_{\Sigma}\phi$ and are surface-divergence free. Thus one geometric direction is not sufficient in general. Two independent tangent directions are the natural generic requirement.

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